Problem 08: AI-Powered Supply Chain Digital Twin for Demand Forecasting and Disruption Recovery

Domain: Operations Research | Time-Series Forecasting | Optimization | Reinforcement Learning

Difficulty: Hard

Problem

Retailers must balance inventory cost, demand, supplier lead times, warehouse capacity and transport delays. Forecasts alone do not say how to redistribute stock when a supplier fails, demand spikes or a warehouse goes down.

Objective

Build a simulation-based decision-support system for a network of suppliers, warehouses and stores that predicts stockouts and recommends recovery strategies.

Users

Supply-chain planners, retail operations managers.

Core Requirements

- Model suppliers, warehouses, products, stores.
- Ingest historical demand and inventory.
- Forecast demand per product-location.
- Model lead times and replenishment.
- Simulate inventory movement.
- Detect stockouts and excess stock.
- Recommend reorder quantities and redistribution.
- Simulate disruptions (supplier failure, transport delay).
- Compare strategies on cost, service level, unmet demand.
- Visualize network state and outcomes.

Advanced Challenges

- Forecasting combined with optimization.
- Uncertain demand and lead times.
- Reorder policies under capacity limits.
- Centralized vs decentralized decisions.
- Discrete-event simulation of disruptions.
- Optional RL replenishment policy.
- Evaluation across stochastic scenarios (Monte Carlo).

Suggested Datasets

M5 Forecasting (Walmart), Favorita Grocery Sales, Olist e-commerce, DataCo Supply Chain; synthetic network topology and lead times.

Suggested Tech

Pandas, LightGBM/XGBoost, OR-Tools or PuLP, SimPy, NetworkX, FastAPI, Streamlit; Stable-Baselines3 for RL extension.

Reference Architecture

Demand forecaster (quantiles) → SimPy twin (network, lead-time distributions) → policy layer ((s,S), newsvendor, MILP redistribution, optional RL) → scenario runner (N Monte Carlo runs) → KPI store → dashboard.

Evaluation

Total operating cost, stockout frequency, fill rate, service level, recovery time, results under disruptions vs baseline policy. Key point: show that forecasts change decisions.

Key Requirement

A forecasting model alone is insufficient; show how predictions drive operational actions.
Pitfalls
Deterministic simulation only; comparing policies on a single scenario.
Suggested Timeline (48h)
0-8h data + forecaster; 8-20h simulator; 20-32h optimizer/policies; 32-40h disruptions + MC eval; 40-48h UI.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work